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CASE STUDY
(SEPM)

Aim to understand devops Principles, Practices & DevOps Engineer Role & Responsibilities

1. Introduction to DevOps

DevOps is a combination of Development (Dev) & Operations (Ops) that focuses on implaying collaboration, automation & the speed of software delivery. It enables organizations to deliver applications at higher velocity while enhancing communication, collaboration & integration between teams

Problems before DevOps

Before the adoption of DevOps, organizations faced several challenges

- slow deployment cycles often taking months instead of days
- program conflicts between development & operation teams
- bugs detected late during testing phase
- inconsistent environments across development, testing production

2. DevOps principles (CALMS framework)

The CALMS framework defines the core principles of DevOps

i) Culture
- emphasizes collaboration between teams
- promotes shared responsibilities for outcomes

ii) Automation
- reduces manual effort
- automates testing, deployment & infrastructure

iii) Lean
- focuses on eliminating waste

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g) Micro services architecture
- applications are divided into smaller services
- improves flexibility & scalability

h) Containerization
- uses containers
- ensures applications run consistently across environments

i) Continuous monitoring
- tracks system performance & health
- helps in early detection of issues

j) Cloud - based DevOps
- utilizes cloud platform for automation & scaling
- enhances flexibility & resource utilization.

4) DevOps Engineer roles & responsibilities
A DevOps engineer acts as a bridge between development & operations team
Key responsibilities:
a) CI/CD pipeline
manages automated integration & deployment pipeline
b) Automation
automates build, test & deployment tasks
c) Infrastructure
manages infrastructure using IaC tools
d) Monitoring & logging
monitors performance & analyzes logs
e) Collaboration
coordinates with development, QA & operations teams
f) security (DevSecOps) Int
Integrates security into pipelines